

Miguel Grisales Coca

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EXPERIENCE

Assistive Robotics Lab — Dr. Aman Behal 10hr/week <i>Undergraduate Research Assistant, University of Central Florida, College of Engineering and Computer Science</i> Design and fabrication of components for Kinova Gen-3 robotic manipulator arm.	Sep. 2025 – Mar. 2026
EMBRACE Lab — Dr. Ladda Thiamwong 30hr/week <i>OPS-Student, Undergraduate Research Assistant, University of Central Florida, College of Nursing</i> - Selected data collection strategies and wrote the standard operating procedures. - Translated instructions and tests for Spanish speaking participants. - Administered strength, mobility, body composition, and cognitive function assessments.	Oct. 2022 – Feb. 2025
WEAR Lab — Dr. Joon-Hyuk Park 30hr/week <i>Undergraduate Research Assistant, University of Central Florida, College of Engineering and Computer Science</i> - Created pilot study to evaluate a wearable actigraphy device. Resulting in NIH-funded grant. - Lead designer on multiple projects using CAD, FEA, 3D printing, and machining. - Developed algorithms for embedded controllers in C/C++.	Nov. 2021 – Apr. 2024

EDUCATION

University of Central Florida, Bachelor of Science in Mechanical Engineering GPA: 3.18 <i>Intelligent Robotic Systems – Minor</i>	August 2026
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SKILLS

- Embedded Controls & Robotics:** ROS2 (Intermediate), Linux Ubuntu CLI (Proficient), NVIDIA Isaac Lab (Intermediate)
- Programming:** C/C++ (Intermediate), Python (Intermediate), MATLAB (Intermediate),
- Design & Fabrication:** SolidWorks (Proficient), ANSYS FEA (Familiar), 3D Printing (Proficient), Machining (Basic)

PROJECTS

Bring ROAR to Life, Capstone Project, University of Central Florida Faculty's Choice award winner, "Hey, Anky!" Physical-AI controlled and voice teleoperated 14-DOF quadrupedal robot dinosaur. <u>Role: Controls Lead — Developed a full machine learning simulation to hardware deployment pipeline in Isaac Lab and ROS2.</u> - Trained a reinforcement learning model in Isaac Lab 3.0 using GPU accelerated parallel simulation with custom rewards. - Validated the control architecture's performance through Isaac Sim + ROS2 sim in loop testing in Linux Ubuntu CLI. - Applied MATLAB's Robot Toolbox to design a ROS2 compatible control backup using an inverse kinematics approach. - Integrated and calibrated IMU, positional, and audio sensors via custom ROS2 nodes, applying low pass filtering to de-noise data. - Determined actuator stiffness and damping by analyzing rosbag step response test data to calibrate Isaac Sim's actuator model. <u>Role: Technical Lead — Kept the team on track toward its stated goals and provided mentorship.</u> - Mentored teammates and classmates regarding programming, controls, and electromechanical design. - Directed task delegation and milestone reporting across interdisciplinary sub-teams to integrate all work. - Conducted system reliability and failure mode analysis (FMEA) with severity/occurrence/detection ratings.	July 2026
Airfoil Inspired Car, EML 3303C Mechanical Engineering Measurements — Role: CAD Designer Miniature car with airfoil inspired design for drag coefficient and flow analysis in a laser vapor wind tunnel. - Analyzed airfoil (SD7062) drag polars to determine the best angle of attack to maximize downthrust-drag coefficient. - Applied SolidWorks to model a car with an airfoil-shaped body for 3D printing.	July 2025
Dynamic Swept Wing, WEAR Lab, University of Central Florida — Role: Systems Integrator Morphing airfoil for submersed flow analysis. User controlled telescopic span and variable sweep angle. - Designed power transmission, vice-locking cable tension mechanism, and controller housing in SolidWorks. - Miniaturized electrical components and controller by modification of and protoboard soldering. - Optimized controls in C/C++ for speed, accuracy, and ease of use.	April 2025

PUBLICATIONS

Choudhury, R., Park, J.-H., Banarjee, C., **Grisales Coca, M.**, Fukuda, D., Xie, R., Stout, J., and Thiamwong, L., **2024**, "Associations between Monitor-Independent Movement Summary (MIMS) and fall risk appraisal combining fear of falling and physiological fall risk in community-dwelling older adults," *Frontiers in Aging*, 5.

LANGUAGES

English	Speaking: Advanced	Writing: Advanced	Reading: Advanced
Spanish	Speaking: Advanced	Writing: Intermediate	Reading: Intermediate